Mahmoud Ahmed Ali Fahmi Mohamed Salem

B.Sc. Computers and Information Sciences (Software Engineering) · Mansoura University, Egypt

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PROFILE

Software engineer and 2024 graduate of the four-year B.Sc. in Computers and Information Sciences (Software Engineering) at Mansoura University — 135 credit hours, taught and examined entirely in English, final result 81.43 % (CGPA 2.91/4.00, German equivalent grade 2.0). 108 of the 135 credit hours are in computer and information science. The undergraduate curriculum provides a continuous mathematical foundation — calculus, linear algebra, discrete structures, numerical analysis, probability theory and statistical methods — alongside algorithms, data structures, databases, operating systems and computer architecture. Graduation project graded A on the application of artificial intelligence to route optimisation. Working as a software engineer since 2024.

EDUCATION

Mansoura University, Egypt — B.Sc. Computers and Information Sciences (Software Engineering) · *Sep 2020 – Jul 2024*

- 135 credit hours (approx. 240 ECTS — four years, eight semesters) · CGPA 2.91/4.00 · total score 81.43 % · German equivalent grade 2.0 („Very Good").
- Degree, transcript and grading system documented by the university; medium of instruction English (MOI letter).
- Graduation project graded A: „Application of Artificial Intelligence for Optimal Route Selection for a Shipping Company".

EXPERIENCE

Zikola — Flutter Developer · Feb 2026 – Jul 2026

- • Developed and maintained cross-platform applications with Flutter/Dart, integrated RESTful APIs, managed app state across devices.
- • Implemented new features to specification and reviewed them for quality before release.

Self-employed — Freelance Software Developer · Jul 2024 – 2026

- • Full lifecycle of client and own products: UI/UX, backend integration, APIs, testing and store deployment.
- • Products: ShipLink (online store with real-time delivery tracking), UniPath, SoundOra, Kalimny (real-time AI voice tutoring).

Military Service · Mar 2025 – Mar 2026

- • Completed one year of mandatory service.

SELECTED PROJECTS

- **Kalimny — AI voice language-learning platform** · **Live Demo** · **Source Code** — Full-duplex real-time voice lessons over WebSocket (Gemini Live API): AudioWorklet capture → 16 kHz PCM → binary frame decode → playback, with interruption handling and automatic session resumption. Next.js 16 (App Router, RSC), strict TypeScript, Supabase Postgres with RLS and atomic SECURITY DEFINER RPCs for the lesson quota, single-use ephemeral API tokens (server-only key), procedural 3D avatar with audio-driven lip-sync, hashed email OTP, deployed on Vercel.
- **ShipLink — e-commerce and delivery platform** (Jul–Dec 2024) · **Live Demo** · **User APK** · **Driver APK** — Single Flutter codebase with flavours for customer/driver/admin apps, real-time order and driver location tracking, Supabase PostgreSQL + Edge Functions, Google Maps, Paymob payments, web deployment. Grew out of the A-graded graduation project on AI-based route selection.
- **UniPath — study-abroad matching and document platform** · **Get it on Google Play** · **Web App** — Ranking engine with weighted percentage scoring, offline-first architecture (Hive, BLoC), AI document generation, Paymob subscriptions with webhook idempotency, bilingual EN/AR RTL.
- **Nexora — AI career intelligence platform** · **Download APK** · **Source Code** · **Web App** — Fuzzy skill matching and weighted scoring for opportunity matching, AI-generated career profiles, PDF generation, offline-first caching, 408+ passing automated tests.

- **SoundOra — Arabic music application** · **Source Code** · **Landing Page** · **Download APK** — Real-time maqam detection, AI taste profiling, synchronised lyrics, offline caching, landing page deployment.

SKILLS

Programming: Dart, TypeScript, Python (ITI training + self-study), OOP, data structures and algorithms, SQL. — Mathematics: calculus, linear algebra, discrete mathematics, probability, statistics.

Mobile & state management: Flutter, MVVM, BLoC/Cubit, Provider, Hive. **Backend & data:** Supabase (PostgreSQL, Auth, Realtime, Edge Functions), Firebase (Firestore, Auth, Cloud Functions), REST APIs.

Networking: TCP/IP, OSI model, network security fundamentals. Web: TypeScript, React, Next.js, Tailwind CSS, HTML, CSS, responsive UI, WebSocket, Three.js (React Three Fiber), GitHub Pages, WASM.

Payments: Stripe, Stripe Connect, Paymob. **Tools:** Git, GitHub, GitLab, Postman, Android Studio, Google Maps SDK.

CERTIFICATES & TRAINING

- Information Technology Institute (ITI) — 1-month Python/Django development training.
- Consulting of Computer and Information Center (CCIC) — 1-month mobile application development with Flutter.

LANGUAGES

- Arabic — native. English — B2 (degree taught and examined in English; MOI letter). German — A2 (in progress).